

1 Introduction

If you have come across the rigorous math papers related to Finite element method or the textbooks by authors Philippe G. Ciarlet, Dietrich Braess, Alexandre Ern and Jean-Luc Guermond or to understand the workings of fenicsx library to solve PDE's using finite element method. Then these notes might direct you towards sufficient basic concepts and materials to get a grasp on the math terminology and theorems (with proofs) for learning abstract Finite element method.

So far, in my research and the answers to this question "Books on mathematical foundation of finite element methods" on scicomp stackexchange, I have discovered two books "Understanding and Implementing the Finite Element Method" by Mark S. Gockenbach and "Introductory Functional analysis with Applications to Boundary Value Problems and Finite Elements" by B. Daya Reddy, both of these books are written, to address the students coming from engineering background (such as myself) without sufficient knowledge in Real analysis and Functional analysis (which are a backbone for understanding PDE's and their weak forms). At the time of writing these notes, I have not finished these books but I have a vague understanding of vector spaces, Lebesgue integration (which is necessary for $L^p(\Omega)$ space) and the Sobolev space ($H^m(\Omega)$). Even these two "easier" books do not cover the foundations in depth. In order to stay true to myself, I have come to realisation that a formal beginning to real analysis and proof writing is necessary before I venture into functional analysis. I have started the books by Paul Zorn "Understanding Real analysis", "Introduction to Real analysis" by Jiří Lebl and youtube playlist by Jeremy Teitelbaum (this teaches from very basic and is based on the book "Book of Proof" by Richard Hammack)

In next coming pages, I will cover superficially some of the concepts and proofs in Real analysis.

2 Real Analysis

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